Knowledge pre-test

Total score 10 points

Participant ID: _____

Part I. Multiple-Choice Questions (5 questions, 1 point each, total 5 points)

1. In a freeway ramp merging area, which of the following most directly reflects the interaction between merging vehicles and mainline vehicles?

A. Number of mainline lanes
B. Acceptable gap between vehicles
C. Speed limit of the road segment
D. Length of the ramp

2. Compared with a standard ramp merging section, the most distinctive feature of a weaving section is:

A. Mainline vehicles always travel at lower speeds
B. Vehicles have bidirectional or multi-directional lane-changing demands within a short distance
C. Vehicles entering from the ramp are no longer affected by mainline traffic
D. There is typically no speed variation in weaving sections

3. In the formation of traffic conflicts, which of the following is most likely to be a direct triggering factor?

A. Mainline vehicles maintain a stable speed
B. Drivers make sudden lane changes near their target position
C. The design speed of the road is relatively high
D. There are many traffic signs along the road segment

4. When the density of mainline traffic increases, what problem are merging vehicles more likely to encounter?

A. Reduced available gaps, making it more difficult to determine a suitable merging opportunity
B. Greater ease in matching the speed of the mainline traffic
C. Reduced impact of merging behavior on the mainline
D. Lane-changing demand typically disappears

5. Which of the following is most appropriate for describing whether speed coordination issues exist in a merging area?

A. Speed difference between mainline vehicles and merging vehicles
B. Variation in longitudinal road gradient
C. Signal timing at intersections
D. Lane width variation

Part II. True/False Questions (2 questions, 1 point each, total 2 points)

6. In a ramp merging area, traffic conflicts will not occur as long as mainline vehicles do not change lanes.

[ ] True [ ] False

7. Conflict risk in weaving sections is typically associated with multiple lane-changing demands within a short distance.

[ ] True [ ] False

Part III. Short Answer Question (1 question, total 3 points)

8. Briefly explain why weaving sections are generally more complex than standard ramp merging sections.

Answer key and scoring criteria

Answer key B B B A A False True

Question 8

Reference Points, answers should include at least one or more of the following:

Multiple vehicles and multi-directional lane-changing demands

Limited space or short operational distance

Stronger interaction and interference among vehicles

Scoring Guidelines

0 points: Irrelevant or incorrect answer

1 point: States that the scenario is “more complex” without explanation

2 points: Identifies one or two key factors

3 points: Mentions core aspects such as multi-directional lane changes, spatial constraints, and intensified vehicle interactions.